Lab Exercise 20- Deploy a Docker Application using AWS ECR and ECS

1. Objective

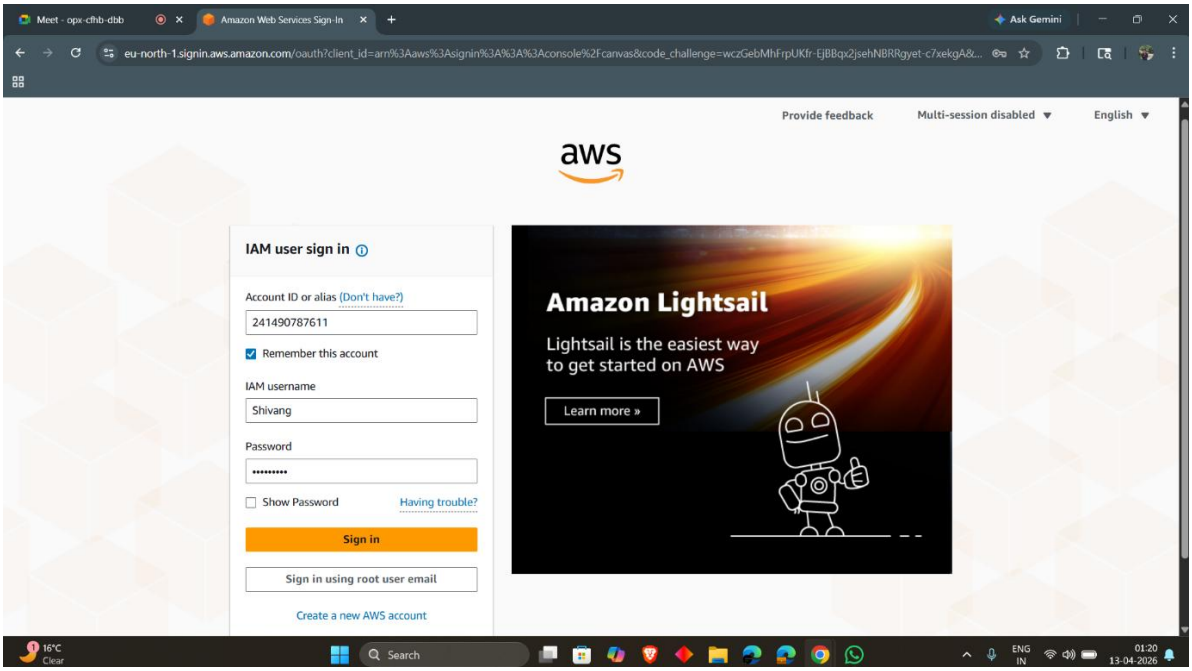
In this lab, you will learn how to:

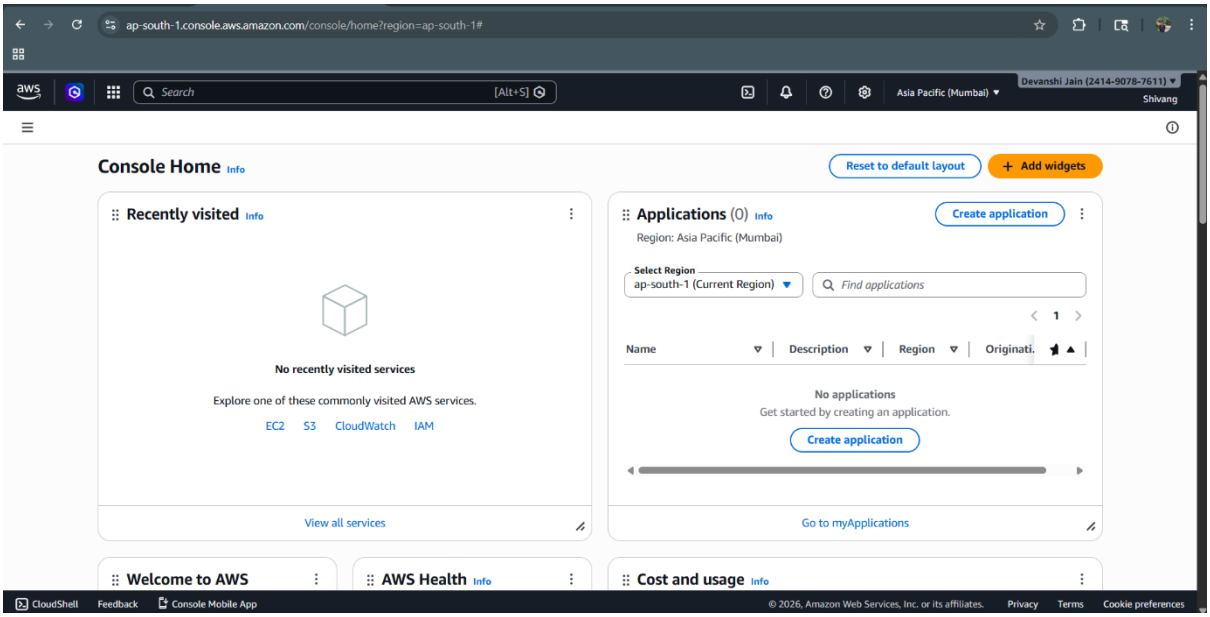
- Build a container image using **Docker**
- Store the container image in **Amazon Elastic Container Registry**
- Deploy the container on **Amazon Elastic Container Service**
- Access the running application through a public endpoint

2. Prerequisites

Before starting the lab, ensure you have:

- AWS account with appropriate permissions
- Docker installed
- AWS Command Line Interface installed and configured
- Basic understanding of containers
- Internet access





3. Lab Scenario

A development team has built a small web application using containers. Your task as a DevOps engineer is to:

1. Containerize the application using **Docker**
2. Store the container image in **Amazon ECR**

3. Deploy the container to **Amazon ECS**
4. Verify the application is running successfully

4. Lab Tasks

Task 1: Create a Docker Application

Create a simple application.

Example

index.html

```
Hello from AWS ECS Lab
```

Name	Date modified	Type	Size
aboutus	23-09-2023 18:08	Microsoft Edge H...	1 KB
Dockerfile	20-08-2023 16:25	File	1 KB
index	23-09-2023 18:08	Microsoft Edge H...	1 KB

Create a **Dockerfile**

```
FROM nginx

COPY index.html /usr/share/nginx/html/index.html
```

Dockerfile

C: > Users > HP > Desktop > COAS_Lab > Lab_9 > my-web-app > Dockerfile

```
FROM httpd
WORKDIR /usr/local/apache2/htdocs/
COPY index.html .
```

Shivang Mangal **500124202** **Batch – 2**

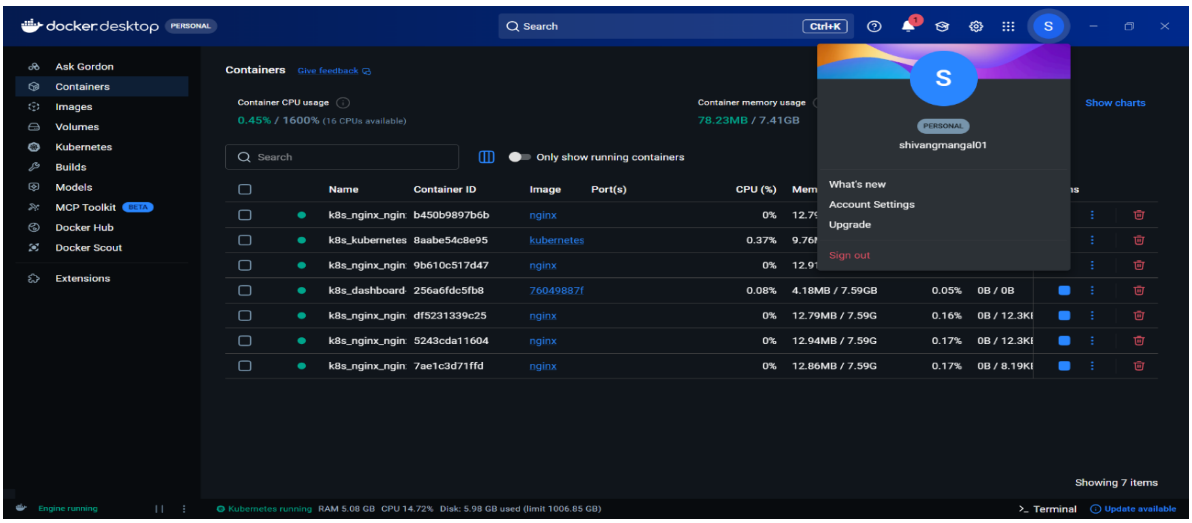
Task 2: Build the Docker Image

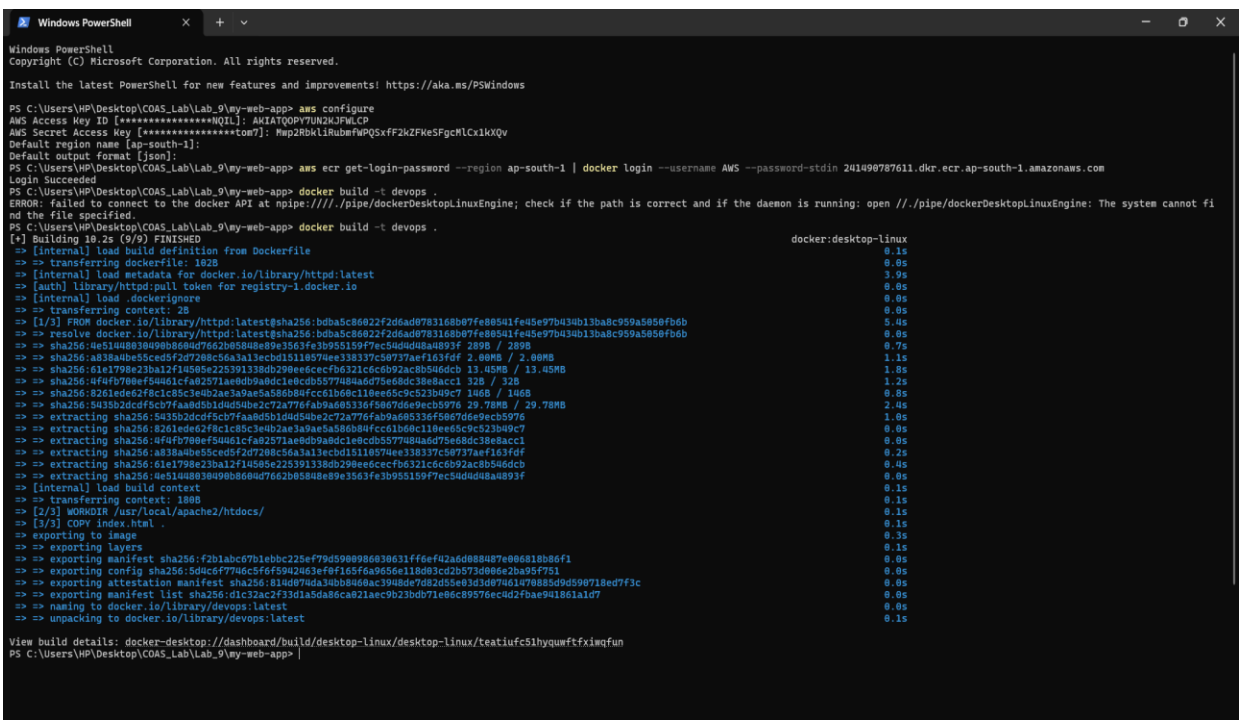
Build the image locally using **Docker**.

```
Docker build -t ecs-lab-app .
```

Verify the image.

```
docker images
```





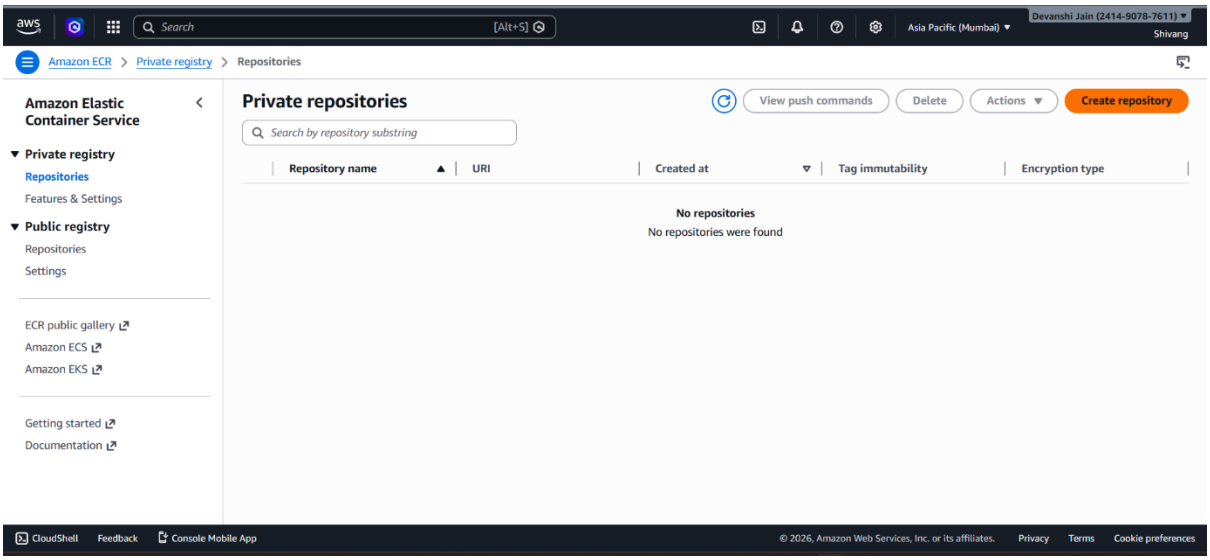
Task 3: Create an ECR Repository

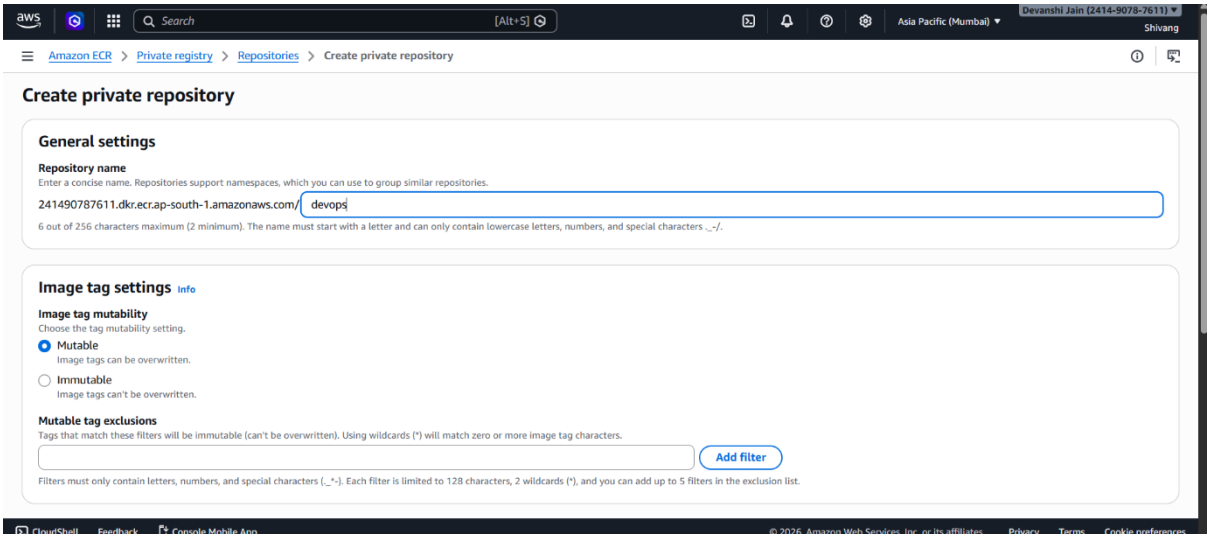
Create a repository in **Amazon ECR**.

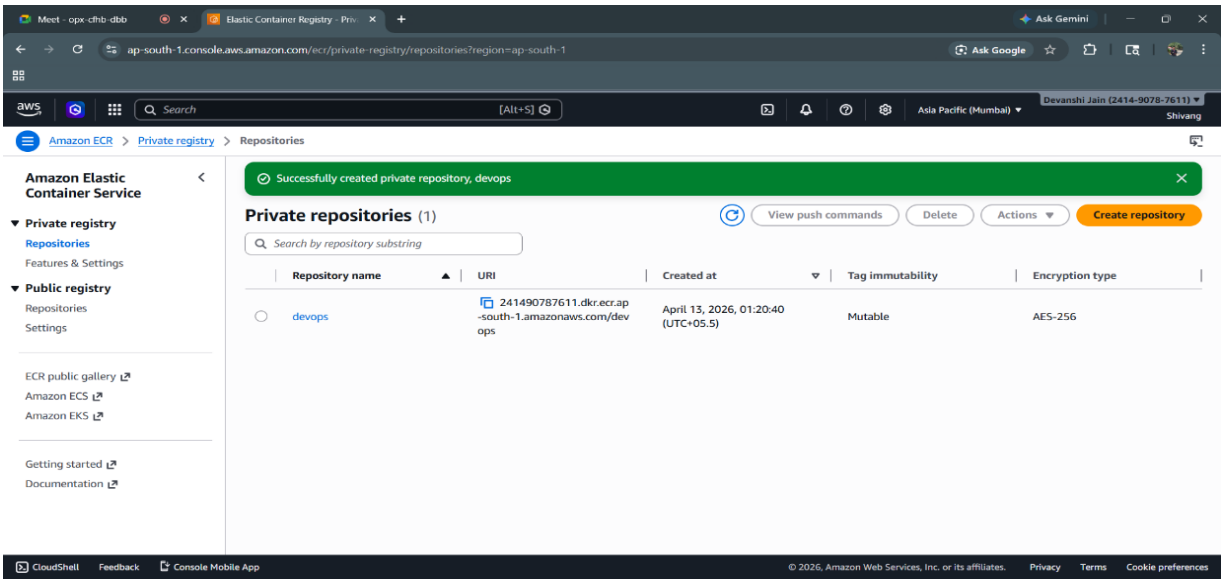
Steps:

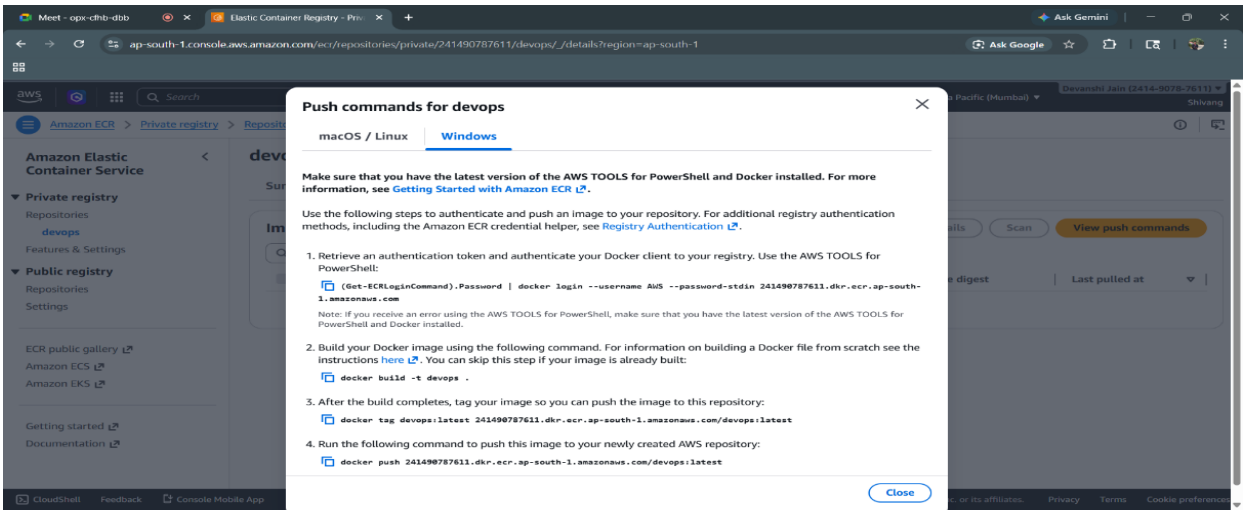
1. Open AWS Console
2. Go to **Amazon ECR**
3. Click **Create Repository**
4. Repository name:

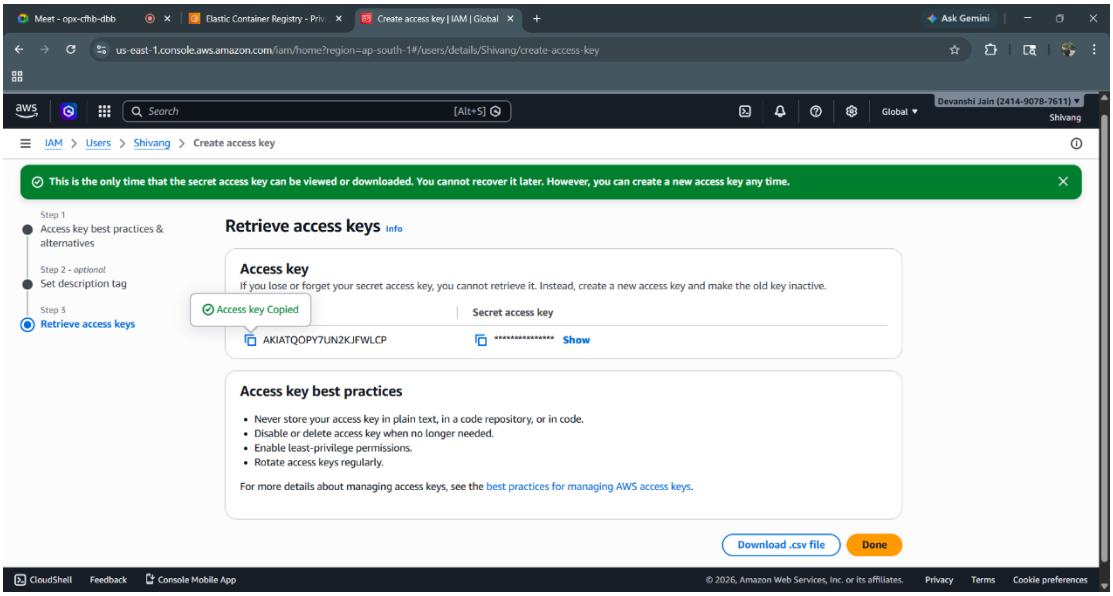
ecs-lab-repository




Amazon ECR > Private registry > Repositories > Create private repository
Create private repository
General settings
Repository name
Enter a concise name. Repositories support namespaces, which you can use to group similar repositories.
241490787611.dkr.ecr.ap-south-1.amazonaws.com/ devops
6 out of 256 characters maximum (2 minimum). The name must start with a letter and can only contain lowercase letters, numbers, and special characters _-/.
Image tag settings Info
Image tag mutability
Choose the tag mutability setting.
Mutable
Image tags can be overwritten.
Immutable
Image tags can't be overwritten.
Mutable tag exclusions
Tags that match these filters will be immutable (can't be overwritten). Using wildcards (*) will match zero or more image tag characters.
Add filter
Filters must only contain letters, numbers, and special characters (._*-). Each filter is limited to 128 characters, 2 wildcards (*), and you can add up to 5 filters in the exclusion list.


Amazon ECR > Private registry > Repositories
Amazon Elastic Container Service
Private registry
Repositories
Features & Settings
Public registry
Repositories
Settings
ECR public gallery
Amazon ECS
Amazon EKS
Getting started
Documentation
Successfully created private repository, devops
Private repositories (1)
View push commands
Delete
Actions
Create repository
Search by repository substring
Repository name
URI
Created at
Tag immutability
Encryption type
devops
241490787611.dkr.ecr.ap-south-1.amazonaws.com/devops
April 13, 2026, 01:20:40 (UTC+05.5)
Mutable
AES-256


Push commands for devops
macOS / Linux
Windows
Make sure that you have the latest version of the AWS TOOLS for PowerShell and Docker installed. For more information, see Getting Started with Amazon ECR.
Use the following steps to authenticate and push an image to your repository. For additional registry authentication methods, including the Amazon ECR credential helper, see Registry Authentication.
1. Retrieve an authentication token and authenticate your Docker client to your registry. Use the AWS TOOLS for PowerShell:
(Get-ECRLoginCommand).Password | docker login --username AWS --password-stdin 241490787611.dkr.ecr.ap-south-1.amazonaws.com
Note: If you receive an error using the AWS TOOLS for PowerShell, make sure that you have the latest version of the AWS TOOLS for PowerShell and Docker installed.
2. Build your Docker image using the following command. For information on building a Docker file from scratch see the instructions here. You can skip this step if your image is already built:
docker build -t devops .
3. After the build completes, tag your image so you can push the image to this repository:
docker tag devops:latest 241490787611.dkr.ecr.ap-south-1.amazonaws.com/devops:latest
4. Run the following command to push this image to your newly created AWS repository:
docker push 241490787611.dkr.ecr.ap-south-1.amazonaws.com/devops:latest
Close

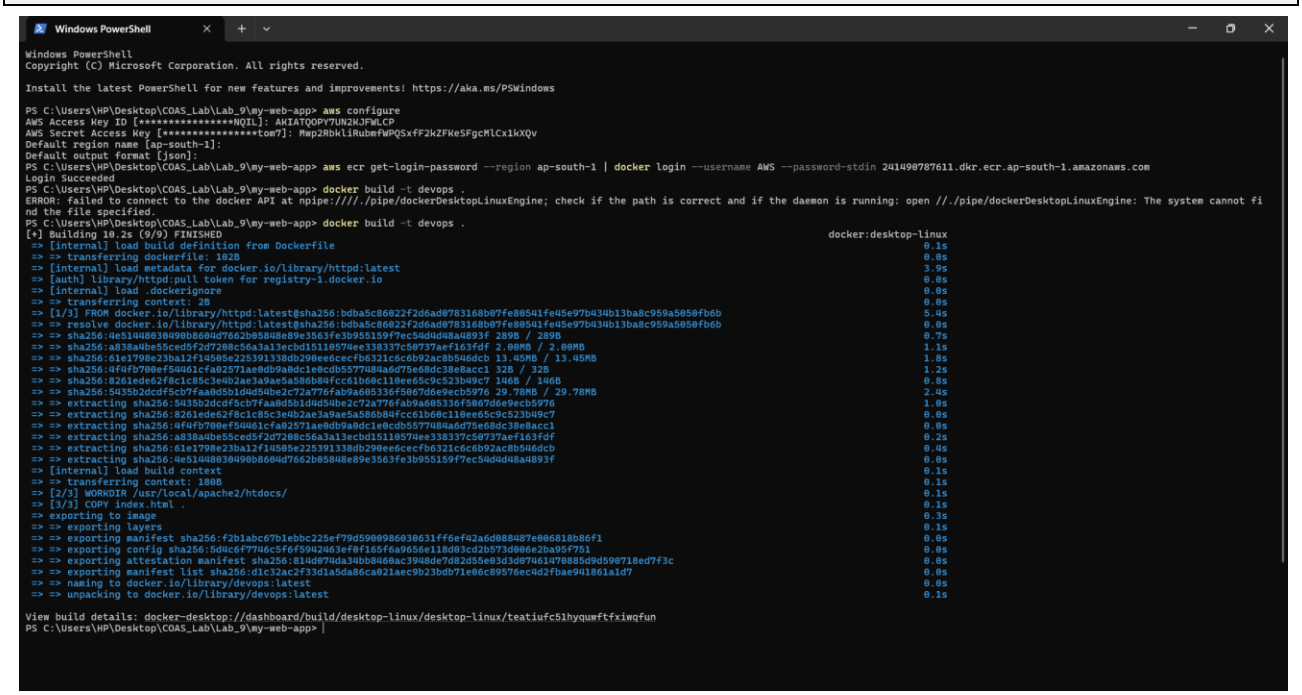


Task 4: Authenticate Docker to ECR

Login to the ECR registry using **AWS CLI**.

Example:

```
aws ecr get-login-password --region us-east-1 | docker login --username AWS --password-stdin
<account-id>.dkr.ecr.us-east-1.amazonaws.com
```



Task 5: Tag the Docker Image

Tag the image to match the ECR repository.

Example:

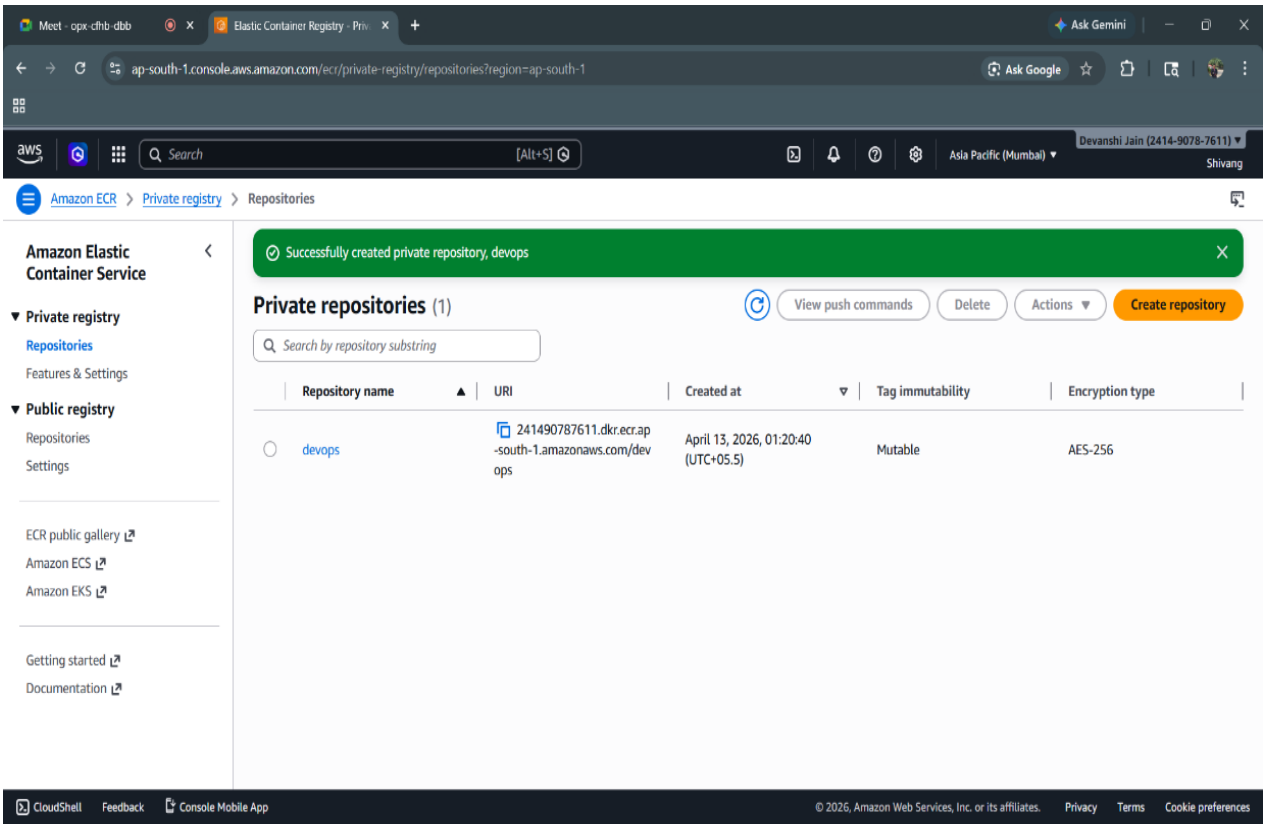
```
docker tag ecs-lab-app:latest <account-id>.dkr.ecr.us-east-1.amazonaws.com/ecs-lab-repository:latest
```

Task 6: Push the Image to ECR

Push the image to the repository.

```
docker push <account-id>.dkr.ecr.us-east-1.amazonaws.com/ecs-lab-repository:latest
```

Verify the image in the **ECR console**.



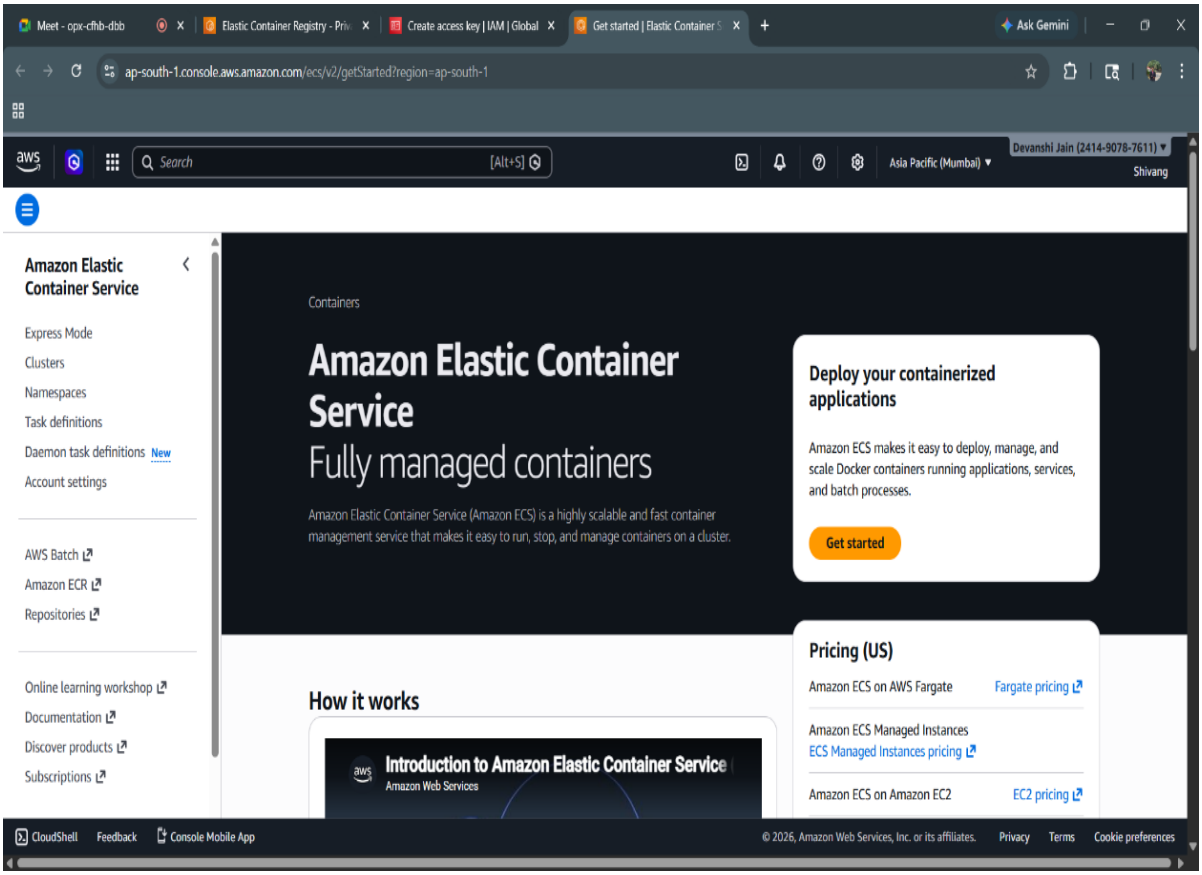
Task 7: Create ECS Cluster

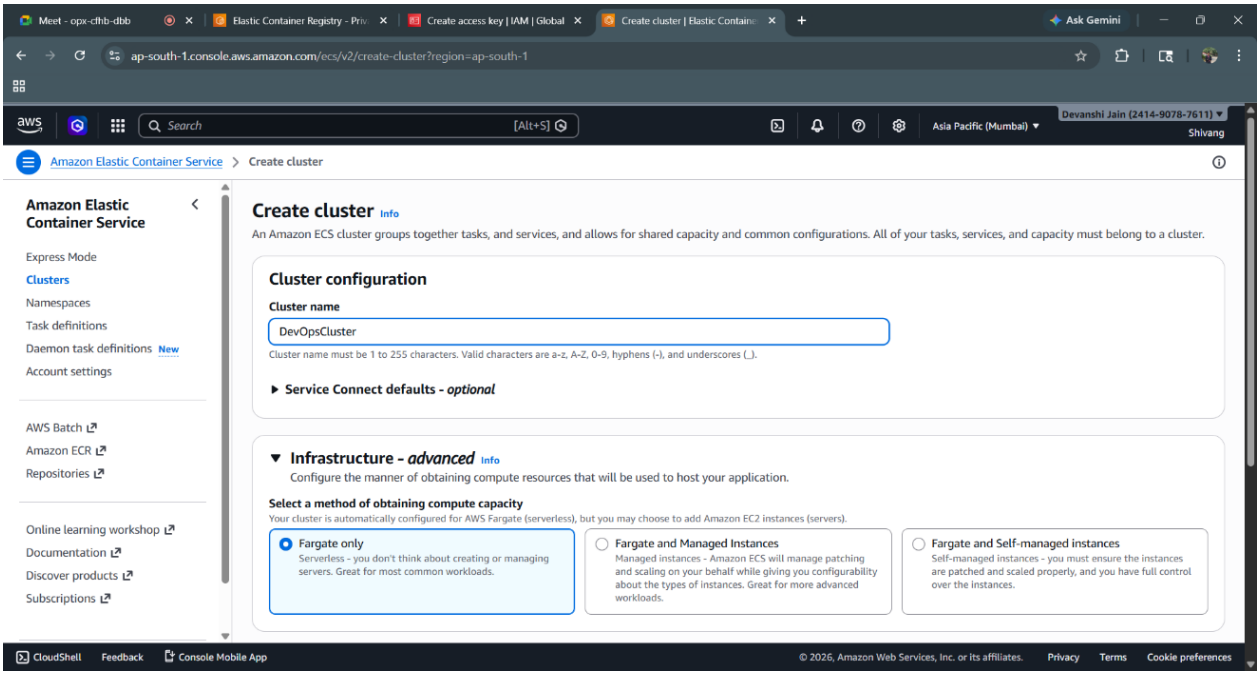
Navigate to **Amazon ECS**.

Steps:

1. Click **Create Cluster**
2. Select **Fargate**
3. Provide cluster name:

ecs-lab-cluster




Meet - opx-cfhb-dbb
Elastic Container Registry - Priv
Create access key | IAM | Global
Create cluster | Elastic Contain
Ask Gemini
ap-south-1.console.aws.amazon.com/ecs/v2/create-cluster?region=ap-south-1
Search [Alt+S]
Asia Pacific (Mumbai)
Devanshi Jain (2414-9078-7611)
Shivang
Amazon Elastic Container Service > Create cluster
Amazon Elastic Container Service
Express Mode
Clusters
Namespaces
Task definitions
Daemon task definitions New
Account settings
AWS Batch
Amazon ECR
Repositories
Online learning workshop
Documentation
Discover products
Subscriptions
Create cluster Info
An Amazon ECS cluster groups together tasks, and services, and allows for shared capacity and common configurations. All of your tasks, services, and capacity must belong to a cluster.
Cluster configuration
Cluster name
DevOpsCluster
Cluster name must be 1 to 255 characters. Valid characters are a-z, A-Z, 0-9, hyphens (-), and underscores (_).
Service Connect defaults - optional
Infrastructure - advanced Info
Configure the manner of obtaining compute resources that will be used to host your application.
Select a method of obtaining compute capacity
Your cluster is automatically configured for AWS Fargate (serverless), but you may choose to add Amazon EC2 instances (servers).
Fargate only
Serverless - you don't think about creating or managing servers. Great for most common workloads.
Fargate and Managed Instances
Managed instances - Amazon ECS will manage patching and scaling on your behalf while giving you configurability about the types of instances. Great for more advanced workloads.
Fargate and Self-managed instances
Self-managed instances - you must ensure the instances are patched and scaled properly, and you have full control over the instances.
CloudShell
Feedback
Console Mobile App
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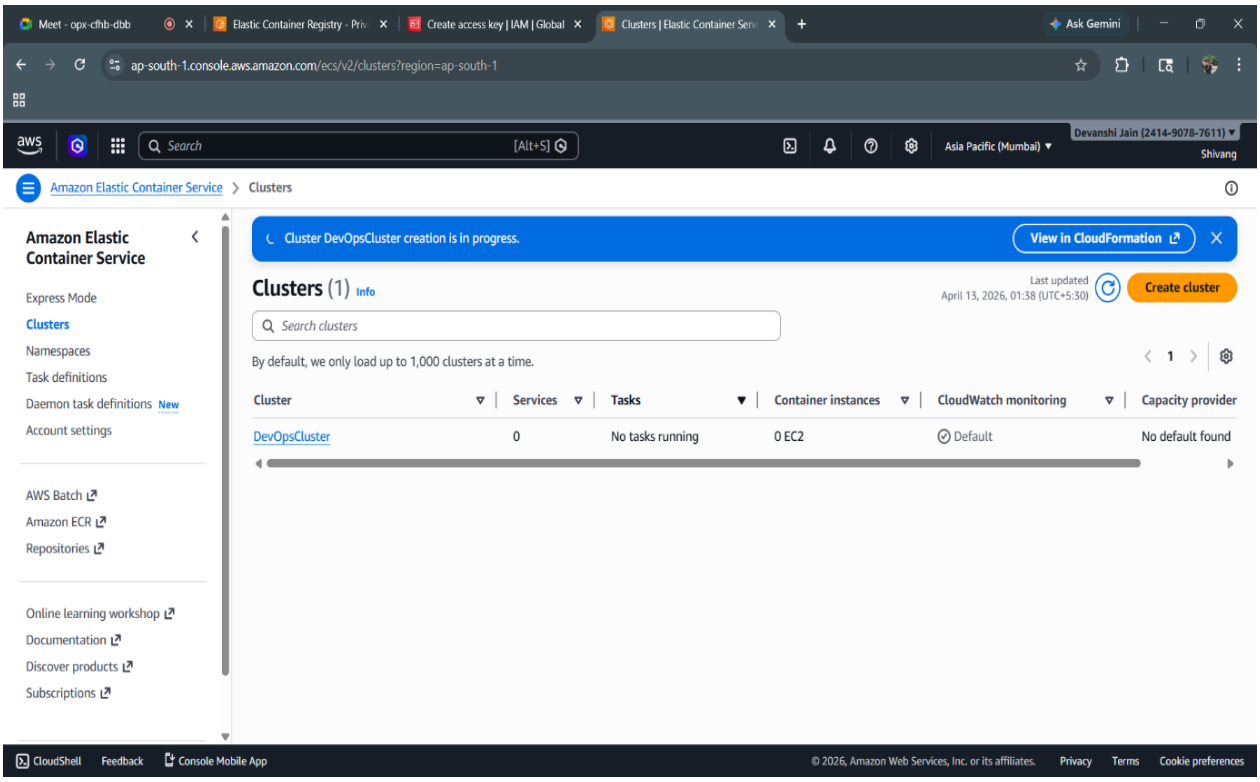
Meet - opx-cfhb-dbb
Elastic Container Registry - Priv
Create access key | IAM | Global
Clusters | Elastic Container Serv
Ask Gemini
ap-south-1.console.aws.amazon.com/ecs/v2/clusters?region=ap-south-1
Search [Alt+S]
Asia Pacific (Mumbai)
Devanshi Jain (2414-9078-7611)
Shivang
Amazon Elastic Container Service > Clusters
Amazon Elastic Container Service
Express Mode
Clusters
Namespaces
Task definitions
Daemon task definitions New
Account settings
AWS Batch
Amazon ECR
Repositories
Online learning workshop
Documentation
Discover products
Subscriptions
Cluster DevOpsCluster creation is in progress.
View in CloudFormation
Clusters (1) Info
Last updated April 13, 2026, 01:38 (UTC+5:30)
Create cluster
Search clusters
By default, we only load up to 1,000 clusters at a time.
Cluster
Services
Tasks
Container instances
CloudWatch monitoring
Capacity provider
DevOpsCluster
0
No tasks running
0 EC2
Default
No default found
CloudShell
Feedback
Console Mobile App
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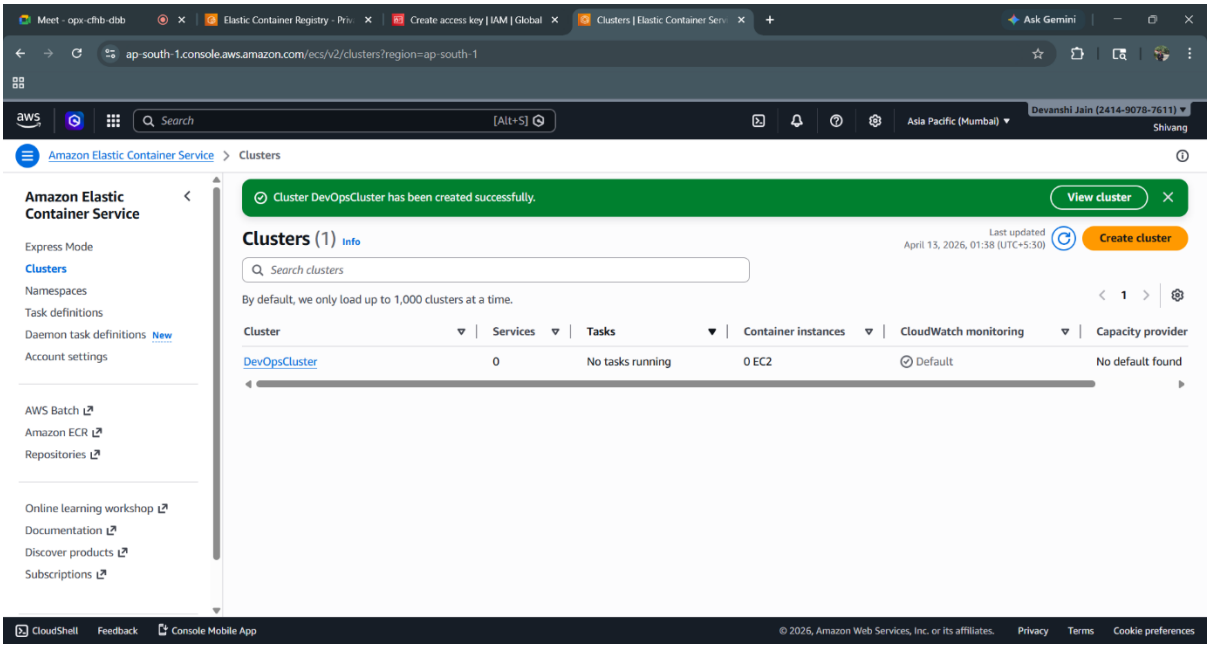


Task 8: Create Task Definition

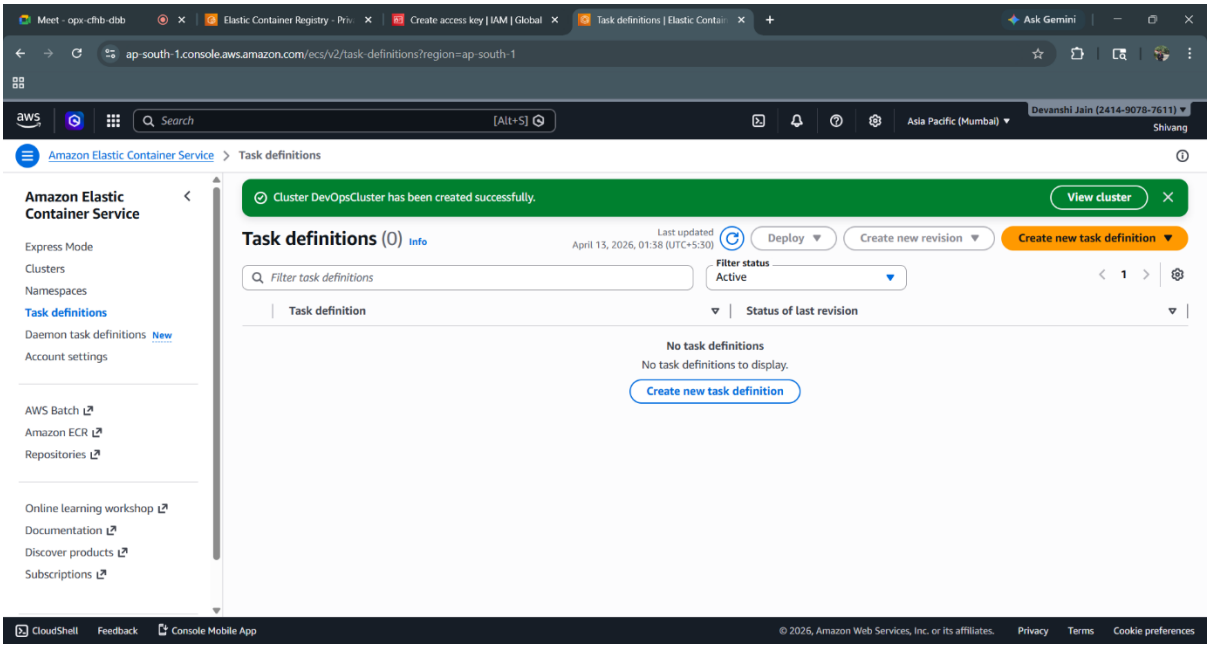
Create a task definition.

Settings:

Container image:

<account-id>.dkr.ecr.us-east-1.amazonaws.com/ecs-lab-repository:latest

Port mapping: 80


Meet - opx-cfhb-dbb
Elastic Container Registry - Priv
Create access key | IAM | Global
Task definitions | Elastic Contain
Ask Gemini
ap-south-1.console.aws.amazon.com/ecs/v2/task-definitions?region=ap-south-1
Search
[Alt+S]
Asia Pacific (Mumbai)
Devanshi Jain (2414-9078-7611)
Shivang
Amazon Elastic Container Service > Task definitions
Amazon Elastic Container Service
Express Mode
Clusters
Namespaces
Task definitions
Daemon task definitions New
Account settings
AWS Batch
Amazon ECR
Repositories
Online learning workshop
Documentation
Discover products
Subscriptions
Cluster DevOpsCluster has been created successfully.
View cluster
Task definitions (0) Info
Last updated April 13, 2026, 01:38 (UTC+5:30)
Deploy
Create new revision
Create new task definition
Filter task definitions
Filter status
Active
Task definition
Status of last revision
No task definitions
No task definitions to display.
Create new task definition
CloudShell
Feedback
Console Mobile App
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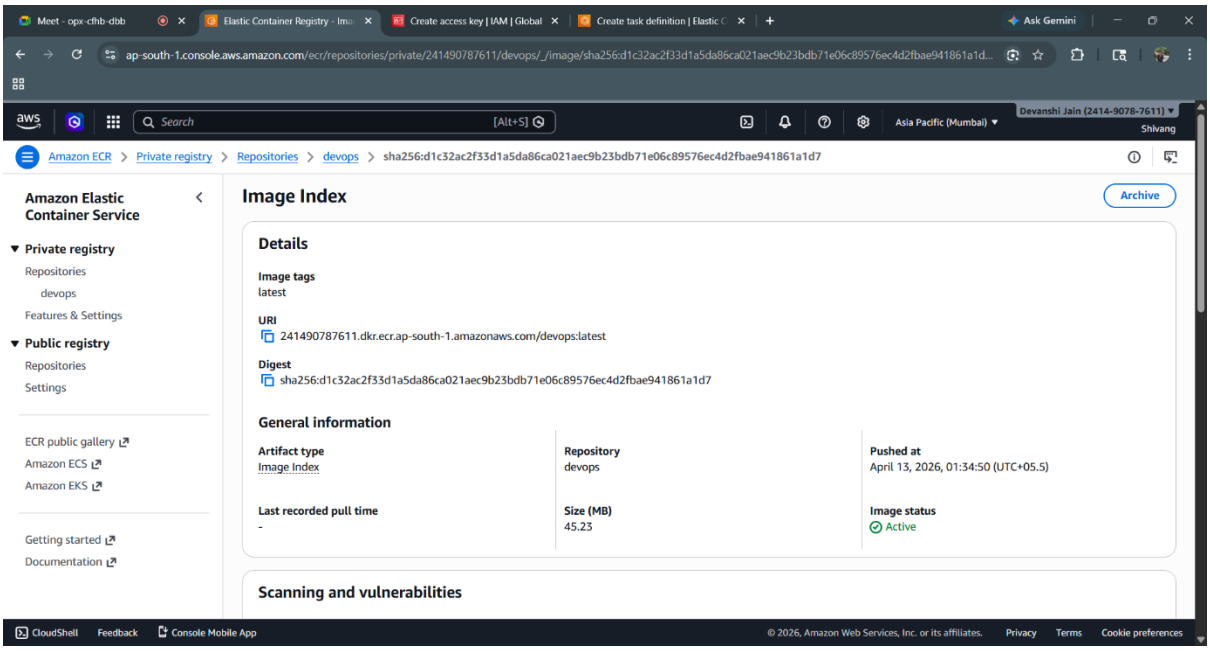
Meet - opx-cfhb-dbb
Elastic Container Registry - Ima
Create access key | IAM | Global
Create task definition | Elastic C
Ask Gemini
ap-south-1.console.aws.amazon.com/ecr/repositories/private/241490787611/devops/_/image/sha256:d1c32ac2f33d1a5da86ca021aec9b23bdb71e06c89576ec4d2fbae941861a1d...
Search
[Alt+S]
Asia Pacific (Mumbai)
Devanshi Jain (2414-9078-7611)
Shivang
Amazon ECR > Private registry > Repositories > devops > sha256:d1c32ac2f33d1a5da86ca021aec9b23bdb71e06c89576ec4d2fbae941861a1d7
Amazon Elastic Container Service
Private registry
Repositories
devops
Features & Settings
Public registry
Repositories
Settings
ECR public gallery
Amazon ECS
Amazon EKS
Getting started
Documentation
Image Index
Archive
Details
Image tags
latest
URI
241490787611.dkr.ecr.ap-south-1.amazonaws.com/devops:latest
Digest
sha256:d1c32ac2f33d1a5da86ca021aec9b23bdb71e06c89576ec4d2fbae941861a1d7
General information
Artifact type
Image Index
Repository
devops
Pushed at
April 13, 2026, 01:34:50 (UTC+05.5)
Last recorded pull time
-
Size (MB)
45.23
Image status
Active
Scanning and vulnerabilities
CloudShell
Feedback
Console Mobile App
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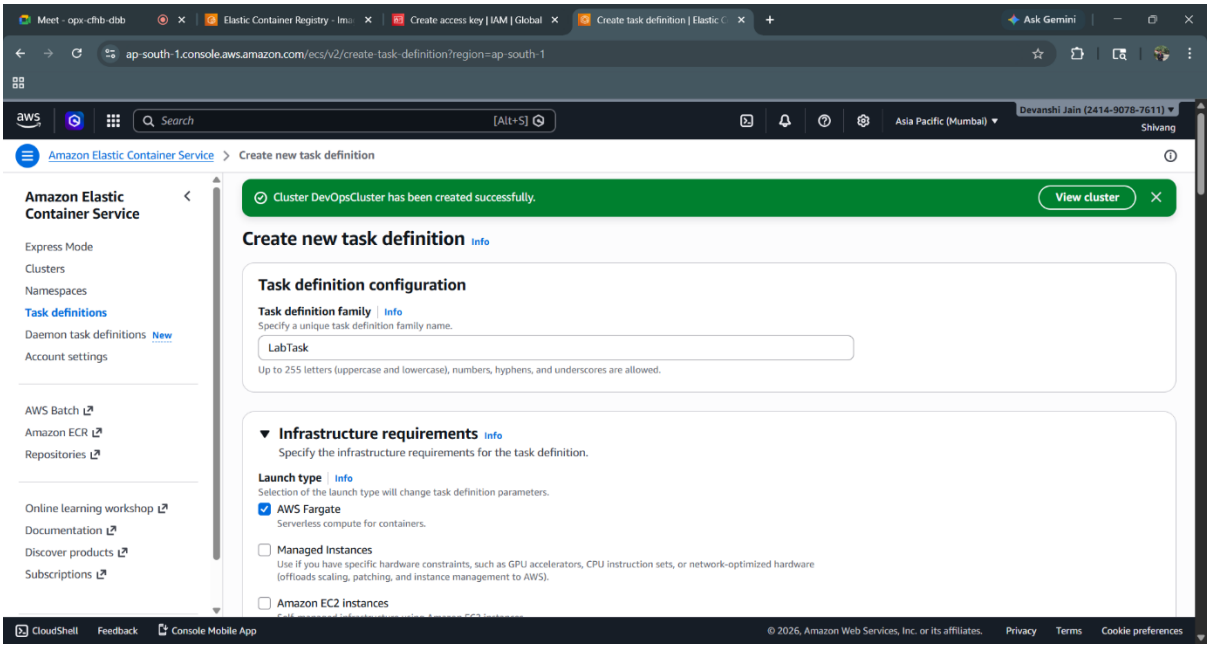
Cookie preferences


Meet - opx-cfhb-dbb
Elastic Container Registry - Ima
Create access key | IAM | Global
Create task definition | Elastic C
Ask Gemini
ap-south-1.console.aws.amazon.com/ecs/v2/create-task-definition?region=ap-south-1
Search
[Alt+S]
Asia Pacific (Mumbai)
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Shivang
Amazon Elastic Container Service
Create new task definition
Amazon Elastic Container Service
Express Mode
Clusters
Namespaces
Task definitions
Daemon task definitions New
Account settings
AWS Batch
Amazon ECR
Repositories
Online learning workshop
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Discover products
Subscriptions
Cluster DevOpsCluster has been created successfully.
View cluster
Create new task definition Info
Task definition configuration
Task definition family Info
Specify a unique task definition family name.
LabTask
Up to 255 letters (uppercase and lowercase), numbers, hyphens, and underscores are allowed.
Infrastructure requirements Info
Specify the infrastructure requirements for the task definition.
Launch type Info
Selection of the launch type will change task definition parameters.
AWS Fargate
Serverless compute for containers.
Managed Instances
Use if you have specific hardware constraints, such as GPU accelerators, CPU instruction sets, or network-optimized hardware (offloads scaling, patching, and instance management to AWS).
Amazon EC2 instances
CloudShell
Feedback
Console Mobile App
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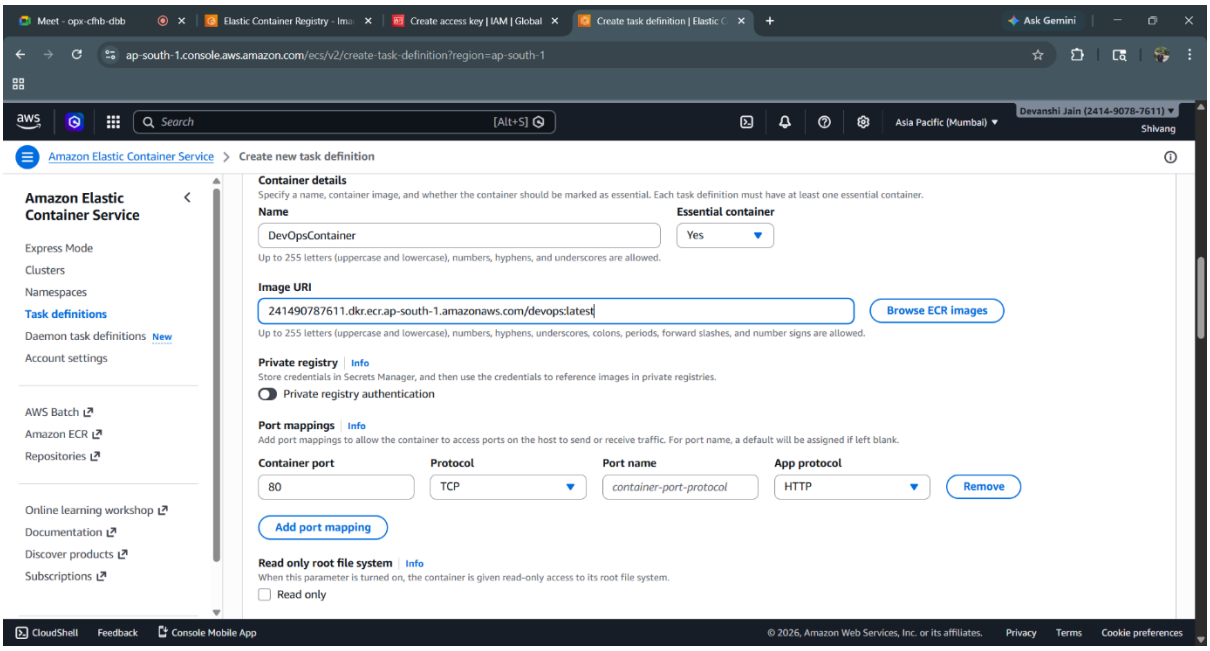
Meet - opx-cfhb-dbb
Elastic Container Registry - Ima
Create access key | IAM | Global
Create task definition | Elastic C
Ask Gemini
ap-south-1.console.aws.amazon.com/ecs/v2/create-task-definition?region=ap-south-1
Search
[Alt+S]
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Shivang
Amazon Elastic Container Service
Create new task definition
Amazon Elastic Container Service
Express Mode
Clusters
Namespaces
Task definitions
Daemon task definitions New
Account settings
AWS Batch
Amazon ECR
Repositories
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Container details
Specify a name, container image, and whether the container should be marked as essential. Each task definition must have at least one essential container.
Name
DevOpsContainer
Up to 255 letters (uppercase and lowercase), numbers, hyphens, and underscores are allowed.
Essential container
Yes
Image URI
241490787611.dkr.ecr.ap-south-1.amazonaws.com/devops:latest
Browse ECR images
Up to 255 letters (uppercase and lowercase), numbers, hyphens, underscores, colons, periods, forward slashes, and number signs are allowed.
Private registry Info
Store credentials in Secrets Manager, and then use the credentials to reference images in private registries.
Private registry authentication
Port mappings Info
Add port mappings to allow the container to access ports on the host to send or receive traffic. For port name, a default will be assigned if left blank.
Container port
80
Protocol
TCP
Port name
container-port-protocol
App protocol
HTTP
Remove
Add port mapping
Read only root file system Info
When this parameter is turned on, the container is given read-only access to its root file system.
Read only
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Console Mobile App
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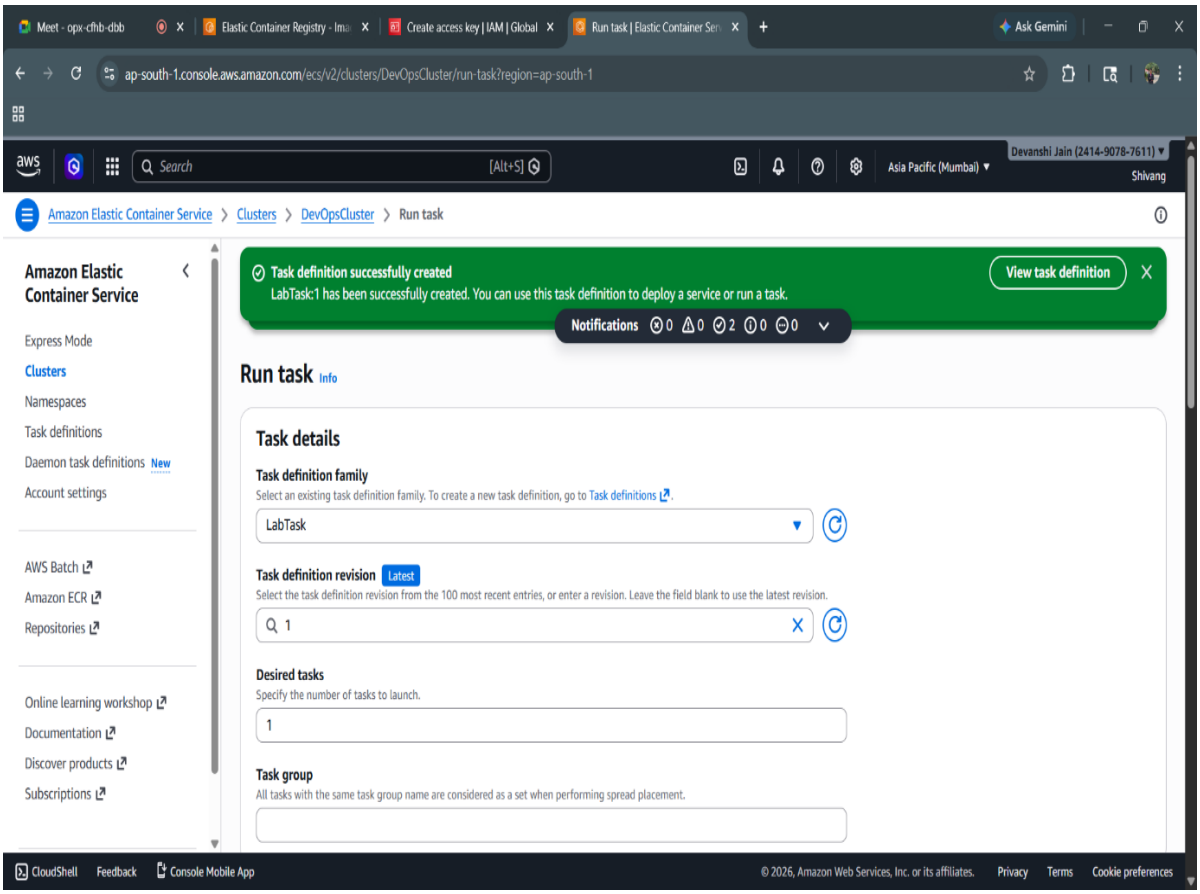
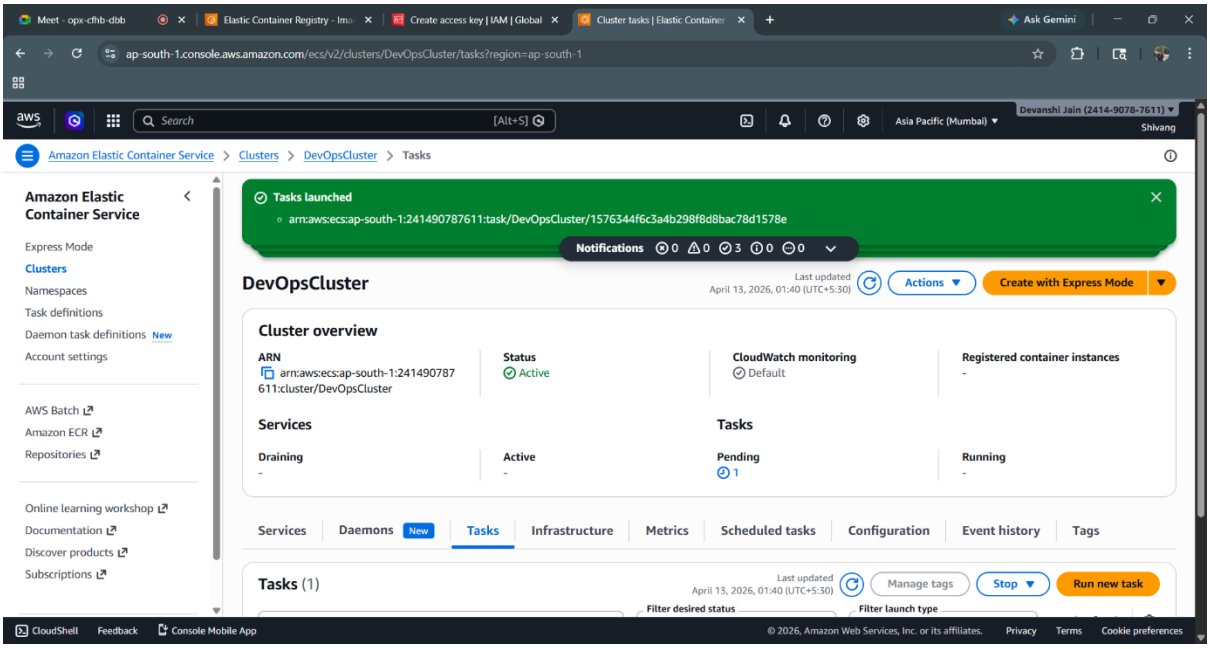
Task 9: Run ECS Service

Deploy the container.

Steps:

1. Create service
2. Select **Fargate**
3. Desired tasks:
4. Enable public IP




Meet - opx-cfhb-dbb
Elastic Container Registry - Ima
Create access key | IAM | Global
Cluster tasks | Elastic Container
Ask Gemini
ap-south-1.console.aws.amazon.com/ecs/v2/clusters/DevOpsCluster/tasks?region=ap-south-1
Search [Alt+S]
Asia Pacific (Mumbai)
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Shivang
Amazon Elastic Container Service > Clusters > DevOpsCluster > Tasks
Amazon Elastic Container Service
Express Mode
Clusters
Namespaces
Task definitions
Daemon task definitions New
Account settings
AWS Batch
Amazon ECR
Repositories
Online learning workshop
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Discover products
Subscriptions
Tasks launched
arn:aws:ecs:ap-south-1:241490787611:task/DevOpsCluster/1576344f6c3a4b298f8d8bac78d1578e
Notifications
DevOpsCluster
Last updated April 13, 2026, 01:40 (UTC+5:30)
Actions
Create with Express Mode
Cluster overview
ARN
arn:aws:ecs:ap-south-1:241490787611:cluster/DevOpsCluster
Status
Active
CloudWatch monitoring
Default
Registered container instances
Services
Draining
Active
Tasks
Pending
1
Running
Services
Daemons New
Tasks
Infrastructure
Metrics
Scheduled tasks
Configuration
Event history
Tags
Tasks (1)
Last updated April 13, 2026, 01:40 (UTC+5:30)
Manage tags
Stop
Run new task
Filter desired status
Filter launch type
CloudShell
Feedback
Console Mobile App
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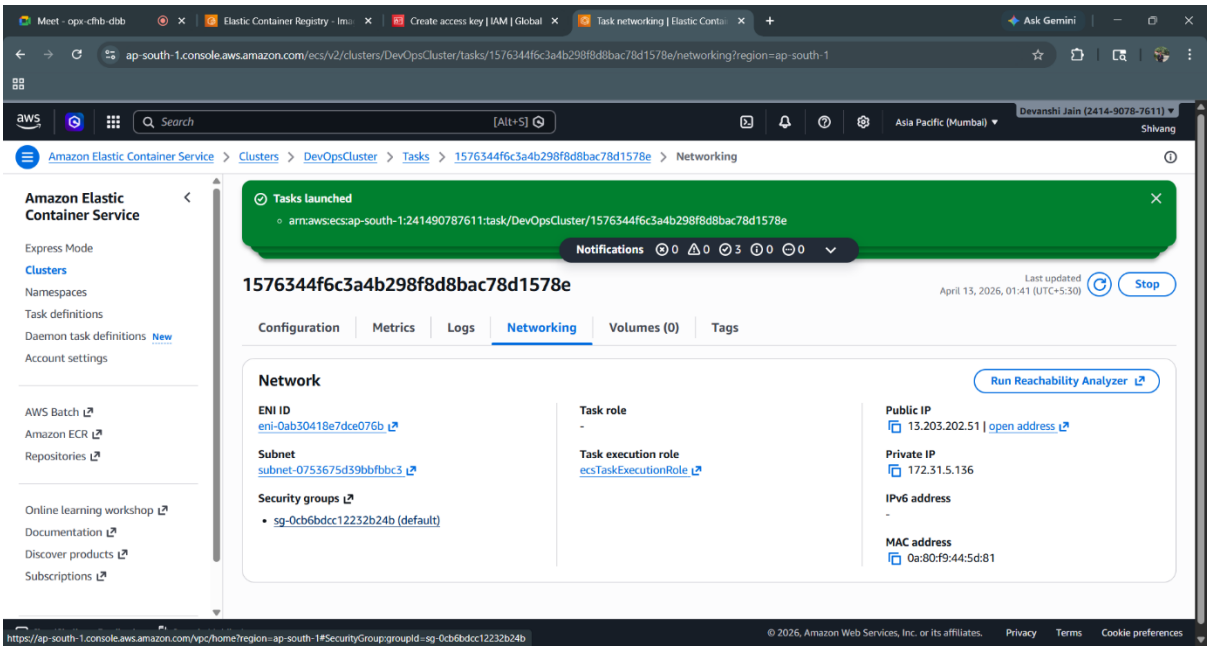
Meet - opx-cfhb-dbb
Elastic Container Registry - Ima
Create access key | IAM | Global
Task networking | Elastic Conta
Ask Gemini
ap-south-1.console.aws.amazon.com/ecs/v2/clusters/DevOpsCluster/tasks/1576344f6c3a4b298f8d8bac78d1578e/networking?region=ap-south-1
Search [Alt+S]
Asia Pacific (Mumbai)
Devanshi Jain (2414-9078-7611)
Shivang
Amazon Elastic Container Service > Clusters > DevOpsCluster > Tasks > 1576344f6c3a4b298f8d8bac78d1578e > Networking
Amazon Elastic Container Service
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Tasks launched
arn:aws:ecs:ap-south-1:241490787611:task/DevOpsCluster/1576344f6c3a4b298f8d8bac78d1578e
Notifications
1576344f6c3a4b298f8d8bac78d1578e
Last updated April 13, 2026, 01:41 (UTC+5:30)
Stop
Configuration
Metrics
Logs
Networking
Volumes (0)
Tags
Network
Run Reachability Analyzer
ENI ID
eni-0ab30418e7dce076b
Subnet
subnet-0753675d39bbfbbc3
Security groups
sg-0cb6bdcc12232b24b (default)
Task role
Task execution role
ecsTaskExecutionRole
Public IP
13.203.202.51 | open address
Private IP
172.31.5.136
IPv6 address
MAC address
0a:80:f9:44:5d:81
https://ap-south-1.console.aws.amazon.com/vpc/home?region=ap-south-1#SecurityGroup:groupId=sg-0cb6bdcc12232b24b
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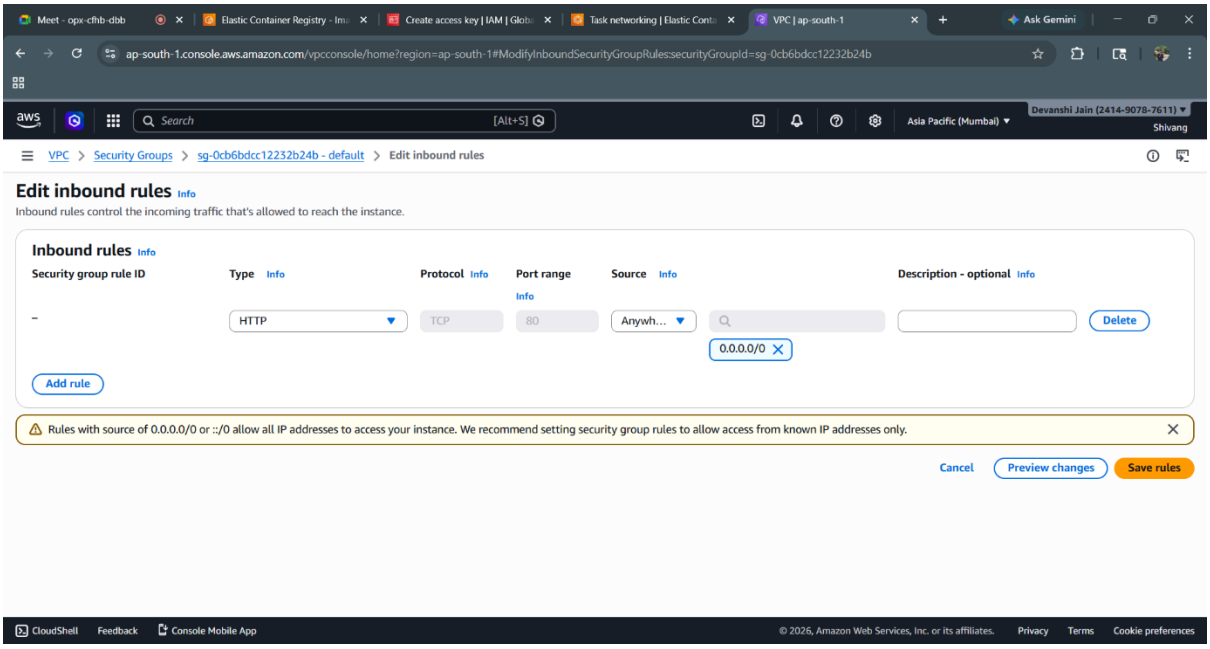
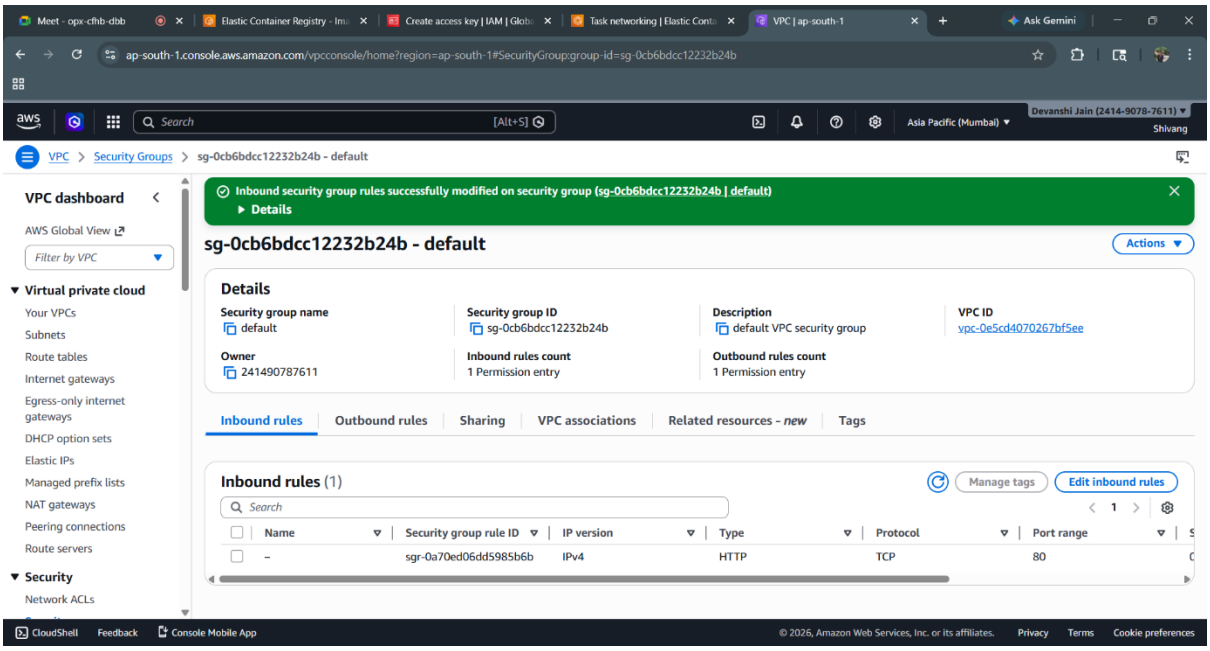




Task 10: Verify Deployment

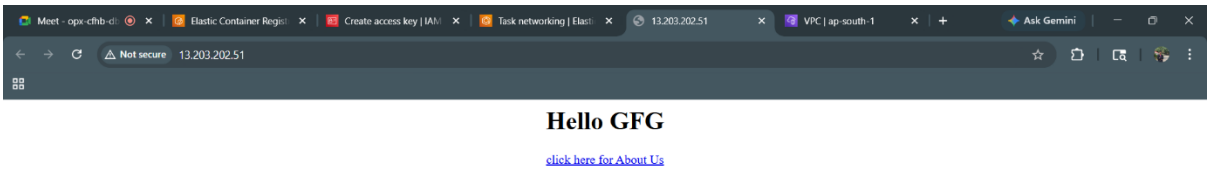
After deployment:

1. Open ECS service
2. Copy the **public IP or load balancer URL**

3. Open in browser

Expected output:

Hello from AWS ECS Lab



Hello GFG

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5. Expected Outcome

After completing this lab:

- Docker image is stored in **Amazon ECR**
- Container is running on **Amazon ECS**
- Application is accessible through a browser